

Build a Smart Sustainable Windhoek: An AR game

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ABSTRACT

The City of Windhoek is dedicated to become a Smart, Sustainable City, in alignment with sustainable development goal 11. Local officials strive to include residents to guide smart and sustainable initiatives and strategies, but lack tools and techniques to engage those unfamiliar with smart city concepts. We proposed a *Build a Smart Sustainable Windhoek* game, which educates and raises awareness while sharing citizen needs and desires to inform smart and sustainable city strategies. It is an interactive hybrid game combining physical interactions with boxes and augmented reality. The game was co-designed with unemployed youth from an informal settlement in the capital. It was played by 8 teams, each consisting of three Windhoek residents from different backgrounds, during a half-day smart city event. The survey results show that participants enjoyed the role of a town planner and building their own city having been teamed up with fellow residents from different suburbs. Moreover, they considered the game to be most informative in creating awareness on Smart City topics.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**.

KEYWORDS

smart city, interaction design, smart technologies, sustainable city, augmented reality, mobile application, resident engagement, city of Windhoek

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1 INTRODUCTION

The City of Windhoek (CoW) is at the forefront of urban development in Namibia and is committed to transforming itself into a Smart, Sustainable, and Caring City. With the use of technology and data, the city aims to improve the quality of life for its residents and promote sustainable transportation, waste management, water

management, among others, in line with SDG 11 (sustainable cities and communities), while also addressing other SDG's.

City officials recognized the importance of involving citizens from different backgrounds and perspectives in the decision-making process to ensure that the city's development is inclusive and equitable for all. By actively engaging with its diverse population, the City of Windhoek is working to create a city that is not only smart and sustainable but also one that prioritizes the needs and well-being of its citizens [2]. However at this point, the City of Windhoek does not have the appropriate tools and techniques to engage residents who have little conceptual understanding and awareness about smart cities, such as inhabitants of the informal settlements. To reduce inequalities (SDG 10), the strategic management team was seeking for inclusive ways to involve a diverse range of citizens in the planning and development of the city. To address this challenge, we collaborated with unemployed youths from Havana Informal settlement to develop an interactive hybrid *Build Your Smart Sustainable City of Windhoek* game. The game aimed to create awareness about smart cities, collect strategic data and create an agentic and enjoyable user experience. The game was piloted with 25 participants as part of a half-day Smart City Hunt event. The game was designed to be understandable by people with little conceptual understanding and awareness about smart cities and at the same time collect residents' preferences and opinions to guide further smart city strategies.

2 RELATED WORK

2.1 Smart cities

Due to the rapid growth of urban population, cities across the globe are faced with enormous challenges that call for flexible and adaptive strategies for urban planning and development. These challenges include dealing with unexpected side effects from mass transit, inadequate urban infrastructure, increasing inequality, unemployment, pollution of the air and water, traffic congestion, and urban crime and violence all contribute to social unrest in cities [3]. To solve some of the difficulties of urbanization and sustainability, smart city concepts were brought forward [8]. Smart City is an urban development concept that aims to securely combine information and communication technology (ICT) and Internet of things (IoT) technologies to manage a city's resources and overcome the challenges of traditional urban development [3]. Building Smart cities requires a relatively new way of thinking about urban planning and design that is focused on the human scale and experience of the city as a whole. In order for a city to be sustainable and have a sustainable urban design, there must be constant and efficient communication between its residents and municipal authorities [12].

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2.2 Residents participation

Residents' participation in the design of a smart city might take many different forms. On a small scale, all residents can be engaged directly, however, if the smart city object of design is going to be used by a well-defined population with common characteristics, representatives from the population can be used as the design participants, as solutions developed by and with citizens promote transparent and participatory collaboration fosters the development and construction of a meaningful living environment [7]. Many Smart City projects declare a wish to involve citizens in their work as crucial players in the city assemblage, acknowledging that technology by itself is insufficient to address problems at the local level. We can argue that citizens have a right to be involved in the design of systems that will affect their daily lives, and incorporating feedback from residents may raise the likelihood that planned smart city projects will satisfy their demands, and promote user approval [5].

Activities such as brainstorming and prototyping, allow citizens to generate as many ideas as possible, making these ideas tangible and visible through drawings, notes, and diagrams [4]. One common prototyping method for facilitating the interaction between the city and its citizens is via Augmented Reality [6]. Augmented reality is useful for prompting interactive user feedback, as it provides an exciting venue for extensive human-machine interactions via physical structure model buildings that can be represented in a real-life interactive object with more details, which often was missing from the traditional urban development[1].

3 METHODOLOGY

Developing an interactive Build a Smart Sustainable Windhoek game was a multi-stage process over three months.

The game was created as a result of numerous engagements with officials from the City of Windhoek, the United Nations Development Programme (UNDP), and the youth from the Havana informal settlement. A co-design approach was used.

The first step in this process was an information session where the City of Windhoek officials outlined their intentions to make Windhoek a smart sustainable city. This session served as an opportunity for us to learn about the city's vision for the future and to provide input on the direction of the project.

The next step was to brainstorm the idea of an interactive Build Your Smart Sustainable City of Windhoek game. The step involved multiple gatherings, collecting input from stakeholders on what features and elements should be included in the game, as well as what challenges and opportunities should be addressed in the present Windhoek.

The next stage was to iteratively develop a prototype of the game. The first version, only consisting of boxes with images, without the AR, was evaluated by the City of Windhoek and UNDP representatives, as well as the youth of Havana (see figure 1).

They were given the platform to play the game and identify which infrastructures they would like to add to the game. Suggestions were made to add buildings such as a church as well as parks and other smart services ensuring citizen safety. Once the most important smart services were identified the youth and the students



Figure 1: Havana youth improving the game

selected videos from youtube that would be suitable and understandable to citizens from different backgrounds. The usability of the AR app was tested by the Havana youth, and feedback was incorporated in the final version.



Figure 2: Boxes with AR triggers

Finally, an evaluation of the game was set-up at the Havana youth cafe as part of a public half-day smart city event in collaboration with the Havana youth, UNDP, a local telecommunication operator MTC, Namibia University of Science and Technology, and the City of Windhoek. 25 invited participants were grouped in teams of 3 citizens from different socio-economic backgrounds. Each team played the game and filled in an online survey consisting of questions about the entire event and in particular about the AR game. This final stage of the real-life evaluation was critical in ensuring that

the game is both effective and inclusive in engaging residents in the process of building a smart sustainable city of Windhoek. Participants provided feedback which is crucial in making the necessary adjustments and improvements to the game to ensure it is effective in engaging residents and promoting sustainable development of the city of Windhoek.

4 THE GAME

4.1 Game mechanism

The game was designed with blocks (boxes) and images of buildings and places in Windhoek, as well as various infrastructures that make up a smart and sustainable city (see figure 2). The main aim of this game is to provide the users with a general overview of the basic concept of a smart and sustainable city. This was realized with AR, whereby participants hover over the images on the boxes to learn more about certain topics. The users were then instructed to place any selected boxes of their choice in the different demarcations on the floor, which were my neighbourhood, city centre, industrial area, and outskirts (see figure 3).



Figure 3: Completed city built

Each group works together to build their own ideal smart city from the provided infrastructure placed on the different boxes.

4.2 Design Objectives

The first objective was to create awareness about a smart and sustainable city. Secondly, to collect data from citizens on their ideal smart city while evaluating what would be implemented first and how they should be implemented. Lastly, create user experiences that are physical, mental, emotional and virtual experiences. Hybrid games, merging digital and physical components, have the potential to merge the best of two worlds, providing new opportunities to enhance gaming experiences [9, 10, 13]. Rogerson et al. [11] have established that hybrid games should promote the interaction among players, draw the players to the board and ensure that technology remains a component only of the board game.

To create awareness the game utilizes an AR application that provides information in the form of videos and audio. This gives



Figure 4: Participants building their city

an opportunity to residents who know little or nothing about a smart and sustainable city to learn about it in a fun and interactive way. Providing information in a visual and auditory format made it easier for the participants to understand and retain the information they are being presented with.

The second objective was achieved by involving the residents in the process of building their own smart city. The data collected from the game will be important in determining what smart city services residents would like to be implemented first and the ones that they consider more important or relevant to them. This will ensure that the smart city services that are implemented are tailored to the specific needs and wants of the residents.

Overall, the game is a unique and innovative way to create awareness about the goal of transforming Windhoek into a smart and sustainable city and to involve the people in the process of building a smart City of Windhoek. The data collected from the game will be of value in determining the best practice of action for the City of Windhoek's smart city initiatives.

4.3 AR Application Development and Installation

The AR application was developed to run on any android device with an android version 8 and above. This application was installed on 10 Samsung A03 android devices. Once the application was launched by the participants, the phone camera was activated in order to scan through the identified AR images presented. There were 12 AR smart city infrastructure images, once the participants hovered over any of the images, a short video about that specific image content pops up and plays. For example, for smart housing, the video displays maximum security, cameras, and fingerprints, and is surrounded by sensors. The topic was selected because home security and safety are a concern for many Windhoek residents. Each and every image scanned has its special content to display to the participant about the infrastructure and how it advantages a smart city (see fig 5).

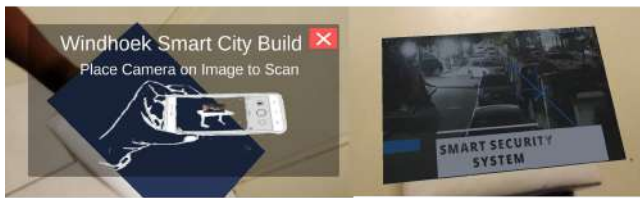


Figure 5: AR Application Interface

5 GAME EVALUATION

A public half-day smart city hunt event was held in November 2022. Our fellow interaction design students created a location-based smart city hunt app which directed participants from one station to another located throughout the city. Our station was hosted in the Havana youth cafe, where teams arrived and played the game. At the end of the smart city hunt experience, each participant filled in an online survey.

5.1 Participants

We had 25 participants grouped into teams of 3 residents, with one being a City of Windhoek or UNDP official, one University student, and one informal settlement inhabitant. The idea was to bring people with different views and understanding to team up to build a smart city together.



Figure 6: Participants

5.2 Participants feedback

All participants of the Smart City Hunt, who filled in the survey (n=24), indicated that they enjoyed the role of a town planner, being in control and were proud of their construction. Furthermore, the participants also indicated that the *Build Your Smart Sustainable City game* was considered most informative in creating awareness on Smart City topics. In addition, participants expressed that they enjoyed building their own city and having been teamed up with fellow citizens from different suburbs. Furthermore, they liked the



Figure 7: Proud Participants

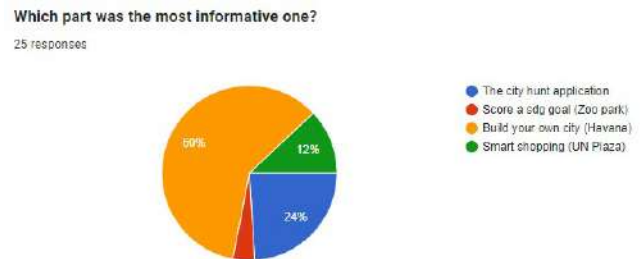


Figure 8: Participants' feedback on awareness

design, creativity, information, placing and locating different elements of services, demarcated locations, and the fact that they can be in charge of their city, which makes them forget their current problems. They felt it was a good experience, educational, gives services to everyone, and opens their mind, being fun and interesting. Participants stated while building, that wifi is a must-have, they need more electrical devices which can reduce carbon emissions which is the first step towards a sustainable environment, they need a stadium, and they need security. Smart city elements are needed in communities, smart technology to consider when building a city, using Augmented Reality, and how to come up with a plan in resource allocation.

5.3 Reflections and way forward

The UNDP and the City of Windhoek considered the half-day event as a pilot to explore the viability of our game for further citizen engagements. While the participants confirmed having had a great user experience, advising further use of the game, we suggest a couple of improvements before scaling its deployment. Firstly, we planned to create our own smart city awareness videos, rather than linking images to youtube videos. Thereby we could create multiple versions in local languages and explain in a more contextually relevant manner. We also intended to use more familiar pictures from the surroundings and show with AR how these places could look

like in the future. In regard to the sizes of boxes, we first intended to create much bigger boxes to play the game in public spaces such as in the central park. However reflecting on a wider distribution, we could create a more portable version with smaller boxes, to be deployed to different institutions, such as NGO's, schools, etc. To enhance awareness on IOT devices, we could integrate smart devices and sensors in the boxes or the game floor. Another challenge has been the collection of meaningful data to inform the cities strategies. While in the interaction the team conversations are informative, the recordings therefore would need to be analysed by specialists in order to be useful.

6 CONCLUSION

We presented an interactive inclusive hybrid game, that was designed to collect data, create awareness and engage residents in building a smart sustainable City of Windhoek. The game can continue to be used by the City of Windhoek to prompt residents' feedback to contribute to urban development strategies. The game broadened the understanding of smart city concepts among the participants, who are now in a position to provide informed suggestions to the city of Windhoek.

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